Task 17: Assignment 17

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Q1. Load the Heart Disease dataset using pandas and display the first 10 rows. Also check the shape and basic information (info()) of the dataset.

```
df = pd.read_excel("heart (1).xlsx")   # Change path if needed

print("First 10 Rows:\n")
print(df.head(10))

print("\nShape of Dataset:")
print(df.shape)

print("\nDataset Information:")
print(df.info())
```

```
First 10 Rows:

   Age Sex ChestPainType  RestingBP  Cholesterol  FastingBS RestingECG  MaxHR  \
0   40   M           ATA        140          289          0     Normal    172
1   49   F           NAP        160          180          0     Normal    156
2   37   M           ATA        130          283          0         ST     98
3   48   F           ASY        138          214          0     Normal    108
4   54   M           NAP        150          195          0     Normal    122
5   39   M           NAP        120          339          0     Normal    170
6   45   F           ATA        130          237          0     Normal    170
7   54   M           ATA        110          208          0     Normal    142
8   37   M           ASY        140          207          0     Normal    130
9   48   F           ATA        120          284          0     Normal    120

  ExerciseAngina  Oldpeak ST_Slope  HeartDisease
0              N      0.0       Up             0
1              N      1.0     Flat             1
2              N      0.0       Up             0
3              Y      1.5     Flat             1
4              N      0.0       Up             0
5              N      0.0       Up             0
6              N      0.0       Up             0
7              N      0.0       Up             0
8              Y      1.5     Flat             1
9              N      0.0       Up             0

Shape of Dataset:
(918, 12)
```

```
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(918, 12)

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 918 entries, 0 to 917
Data columns (total 12 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Age             918 non-null    int64
 1   Sex             918 non-null    object
 2   ChestPainType   918 non-null    object
 3   RestingBP       918 non-null    int64
 4   Cholesterol     918 non-null    int64
 5   FastingBS       918 non-null    int64
 6   RestingECG      918 non-null    object
 7   MaxHR           918 non-null    int64
 8   ExerciseAngina  918 non-null    object
 9   Oldpeak         918 non-null    float64
 10  ST_Slope        918 non-null    object
 11  HeartDisease    918 non-null    int64
dtypes: float64(1), int64(6), object(5)
memory usage: 86.2+ KB
None
```

Q2. Check for missing values in the dataset. Show the count of null values for each column.

```python
print("\nMissing Values:")
print(df.isnull().sum())
```

```
Missing Values:
Age               0
Sex               0
ChestPainType     0
RestingBP         0
Cholesterol       0
FastingBS         0
RestingECG        0
MaxHR             0
ExerciseAngina    0
Oldpeak           0
ST_Slope          0
HeartDisease      0
dtype: int64
```

Q3. Check for duplicate rows in the dataset. If any duplicates are found, remove them and print the new shape of the dataset.

```python
duplicates = df.duplicated().sum()

print("\nDuplicate Rows:", duplicates)

if duplicates > 0:
    df = df.drop_duplicates()
    print("Duplicates removed.")

print("New Shape:", df.shape)
```

```
Duplicate Rows: 0
New Shape: (918, 12)
```

Q4. Identify unrealistic/invalid values: Count how many rows have Cholesterol = 0. Count how many rows have RestingBP 0. Print both counts.

```python
chol_zero = (df["Cholesterol"] == 0).sum()
bp_zero = (df["RestingBP"] == 0).sum()

print("\nRows with Cholesterol = 0 :", chol_zero)
print("Rows with RestingBP = 0 :", bp_zero)
```

```
Rows with Cholesterol = 0 : 172
Rows with RestingBP = 0 : 1
```

Q5. Clean the invalid values: Replace Cholesterol = 0 with the mean cholesterol value (excluding zeros). Replace RestingBP 0 with the mean resting blood pressure value (excluding zeros). Round both columns to 2 decimal places. Print the statistical summary (describe()) of these two columns before and after cleaning.

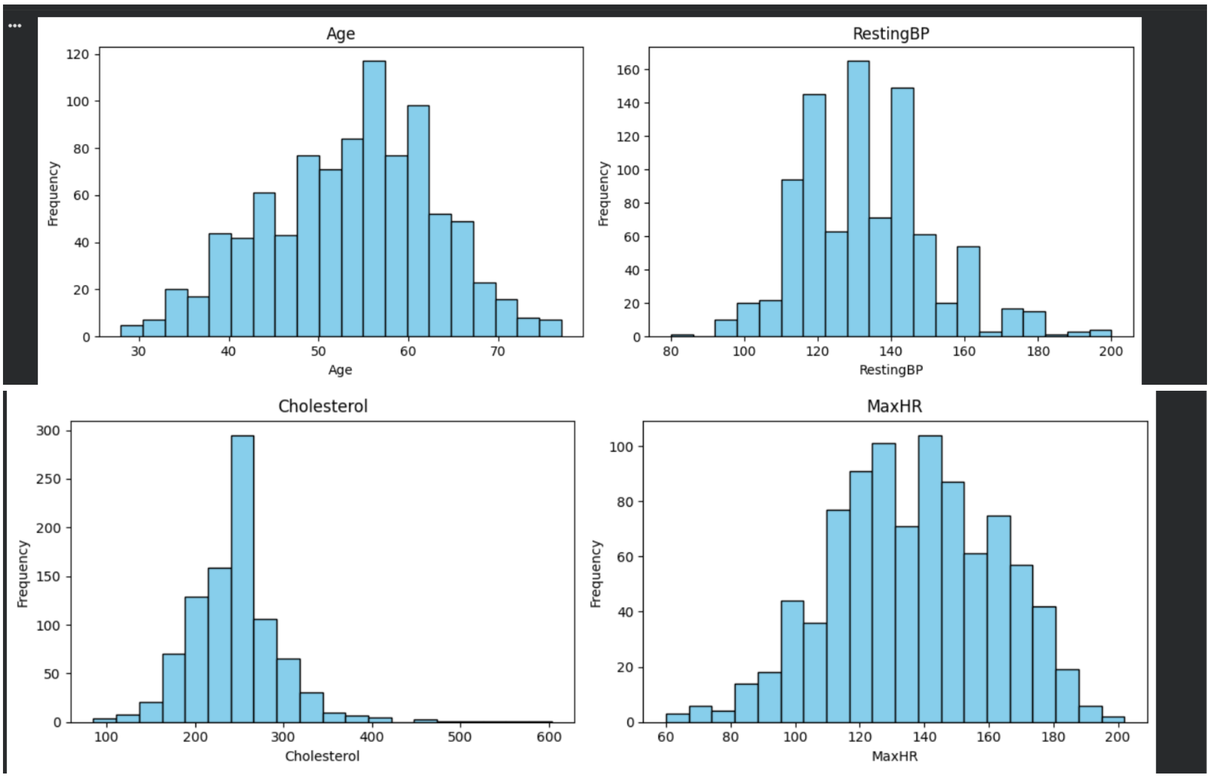
```python
print("\nBefore Cleaning:\n")
print(df[["Cholesterol", "RestingBP"]].describe())

# Mean excluding zeros
chol_mean = df.loc[df["Cholesterol"] != 0, "Cholesterol"].mean()
bp_mean = df.loc[df["RestingBP"] != 0, "RestingBP"].mean()

# Replace zeros
df["Cholesterol"] = df["Cholesterol"].replace(0, chol_mean)
df["RestingBP"] = df["RestingBP"].replace(0, bp_mean)

# Round values
df["Cholesterol"] = df["Cholesterol"].round(2)
df["RestingBP"] = df["RestingBP"].round(2)

print("\nAfter Cleaning:\n")
print(df[["Cholesterol", "RestingBP"]].describe())
```

```
Before Cleaning:

       Cholesterol   RestingBP
count   918.000000  918.000000
mean    198.799564  132.396514
std     109.384145   18.514154
min       0.000000    0.000000
25%     173.250000  120.000000
50%     223.000000  130.000000
75%     267.000000  140.000000
max     603.000000  200.000000
```

```
After Cleaning:

       Cholesterol   RestingBP
count   918.000000  918.000000
mean    244.636253  132.540893
std      53.318029   17.989932
min      85.000000   80.000000
25%     214.000000  120.000000
50%     244.640000  130.000000
75%     267.000000  140.000000
max     603.000000  200.000000
```

Q6. Create a function to plot histograms for the following numerical columns: Age RestingBP Cholesterol MaxHR Plot all four histograms in one figure using subplots 2 2 layout). Use this function to visualize the data after cleaning.

```python
import matplotlib.pyplot as plt

def plot_histograms(data, columns):

    plt.figure(figsize=(12,8))

    for i, col in enumerate(columns, 1):
        plt.subplot(2,2,i)
        plt.hist(data[col], bins=20, color="skyblue", edgecolor="black")
        plt.title(col)
        plt.xlabel(col)
        plt.ylabel("Frequency")

    plt.tight_layout()
    plt.show()

columns = ["Age", "RestingBP", "Cholesterol", "MaxHR"]

plot_histograms(df, columns)
```



Q7. Identify and print numerical columns and categorical columns separatel.

```
import numpy as np

numerical_cols = df.select_dtypes(include=np.number).columns.tolist()

categorical_cols = df.select_dtypes(include="object").columns.tolist()

print("\nNumerical Columns:")
print(numerical_cols)

print("\nCategorical Columns:")
print(categorical_cols)
```

```
Numerical Columns:
['Age', 'RestingBP', 'Cholesterol', 'FastingBS', 'MaxHR', 'Oldpeak', 'HeartDisease']

Categorical Columns:
['Sex', 'ChestPainType', 'RestingECG', 'ExerciseAngina', 'ST_Slope']
```

Q8. Perform One-Hot Encoding on all categorical columns using pd.get_dummies(). Store the result in a new dataframe called df_encoded. Print the shape and the first 5 rows of the encoded dataframe.

```
df_encoded = pd.get_dummies(df, columns=categorical_cols)

print("\nEncoded Data Shape:")
print(df_encoded.shape)

print("\nFirst 5 Rows:")
print(df_encoded.head())
```

```
Encoded Data Shape:
(918, 21)

First 5 Rows:
   Age  RestingBP  Cholesterol  FastingBS  MaxHR  Oldpeak  HeartDisease  \
0   40      140.0        289.0          0    172      0.0             0
1   49      160.0        180.0          0    156      1.0             1
2   37      130.0        283.0          0     98      0.0             0
3   48      138.0        214.0          0    108      1.5             1
4   54      150.0        195.0          0    122      0.0             0

   Sex_F  Sex_M  ChestPainType_ASY  ...  ChestPainType_NAP  ChestPainType_TA  \
0  False   True              False  ...              False             False
1   True  False              False  ...               True             False
2  False   True              False  ...              False             False
3   True  False               True  ...              False             False
4  False   True              False  ...               True             False

   RestingECG_LVH  RestingECG_Normal  RestingECG_ST  ExerciseAngina_N  \
0           False               True          False              True
1           False               True          False              True
2           False              False           True              True
3           False               True          False             False
4           False               True          False              True

   ExerciseAngina_Y  ST_Slope_Down  ST_Slope_Flat  ST_Slope_Up
0             False          False          False         True
1             False          False           True        False
2             False          False          False         True
3              True          False           True        False
4             False          False          False         True

[5 rows x 21 columns]
```

Q9. After cleaning and encoding: Print the final shape of df_encoded. Show the list of all column names in the final dataframe.

```python
print("\nFinal Shape:")
print(df_encoded.shape)

print("\nColumn Names:")
print(df_encoded.columns.tolist())
```

```
Final Shape:
(918, 21)

Column Names:
['Age', 'RestingBP', 'Cholesterol', 'FastingBS', 'MaxHR', 'Oldpeak', 'HeartDisease', 'Sex_F', 'Sex_M', 'ChestPainType_ASY', 'ChestPainType_ATA', 'ChestPainType_NAP', 'ChestPa
```

Q10. Summary) Write a short summary (in comments or markdown) covering: What invalid values did you find and how did you fix them? Why is it important to handle invalid values like Cholesterol = 0? What is the purpose of One-Hot Encoding? Any other observations from the cleaning process.

short summary :

Invalid values found:

- Cholesterol = 0

- RestingBP = 0

Cleaning:

- Replaced zero values with the mean of the respective column

(excluding zeros) and rounded to 2 decimal places.

Importance:

- Invalid values can distort analysis and reduce model accuracy.

- Cleaning improves the quality and reliability of the dataset.

One-Hot Encoding:

- Converts categorical variables into numerical columns so that

machine learning algorithms can use them.

Other observations:

- Missing values and duplicate rows were checked.

- Histograms were created for numerical columns.

- The cleaned and encoded dataset is ready for further analysis.